

Why does therapeutic drug development take so long? Harnessing the immune system to treat cancer: Cancer immunotherapy (part 4)

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This is the last of four case histories related to cancer treatments. Since well before I first started in science, there were discussions about the role of the immune system in combating cancer. The belief, then with limited experimental evidence, was that we all develop abnormal cells starting on paths on the way to cancer all the time but that our immune systems recognize and remove these. Studies of basic immunology, starting in the 1990s, discovered hints of a new immune system regulatory pathway. Over the next decades, details of this pathway were filled in which led to the development of a completely novel approach to cancer treatment based on monoclonal antibody technology introduced in the context of Herceptin in the previous case history. This cancer immunotherapy in dramatically changing treatment of many forms of cancer, including those that were previous intractable.

The immune system performs some truly astounding tasks. It is capable of recognizing and combating infectious agents, even those that not previously encountered. At the same time, it generally avoids attacking cells and tissues in our bodies that are recognized as "self". For decades, scientists have sought to use the immune system to treat cancer. Indeed, our immune systems normally rid us of many developing cancers, as revealed by the fact that immunodeficiency leads to increased cancer development. But how could the immune system be harnessed to address cancers that have already developed?

For many years, the approach was to try to stimulate the immune system to attack known cancers. Cancers do produce and display to the immune systems molecules that are not present in normal tissues. These tumor antigens may be the products of genes that have mutated in tumors or proteins that are normally present only inside cells that are exposed in tumors. Many attempts did lead to some limited successes, but, overall, this promising approach was disappointing. However, as our understanding of basic immunology advanced, a complementary approach was conceived that has proven to be remarkably productive.

The adaptive immune system includes two major classes of cells: B cells, which develop in the spleen, and lymph nodes that produce antibodies, and T cells, which develop in the thymus, some of which can attack cells containing foreign markers. T cells express variable receptors on their surfaces that are capable of recognizing a wide range of foreign molecules. When a T cell encounters a cell that appears to harbor something foreign, it is stimulated to divide to make more identical T cells. However, in addition to this stimulatory signal, scientists discovered that there are also “immune checkpoint” systems that modulate the stimulation. One system involves a protein called “Programmed Cell Death Protein-1” or “PD-1” which is present on the surfaces of T cells and “Programmed Death Ligand-1” or “PD-L1” which is present on the surfaces of some cells. PD-1 was discovered in 1992 and PD-L1 was discovered in 2000 after a long search. When PD-1 engages with PD-L1, this inhibits T cell proliferation and suppresses the immune response. In essence, expressing PD-L1 makes a cell less susceptible to the T cell-mediated immune response. Since tumor cells are selected for their ability to survive, many tumor cells express PD-L1 with estimates of 15% overall but much higher rates for certain types of cancer. This presents an alternative therapeutic strategy. If cancer cells are actively suppressing the immune response to themselves, then blocking the inhibitory pathway might enable the immune system to attack the tumor without the need for any additional stimulation.

The approach used with Herceptin provided a potentially straightforward path to implementation. Monoclonal antibodies were generated against PD-1 with the idea that this antibody would prevent PD-1 from binding PD-L1 and, hence, would prevent the inhibition of the immune response. Antibodies were generated and clinical trials were initiated by 2006. The first treatment was approved in 2014 for metastatic melanoma using the antibody **pembrolizumab (Keytruda)**. Other clinical trials led to approvals for other cancers and, in 2020, the FDA approved **pembrolizumab for any cancer that showed indications of high rates of mutated protein generation, allowing it to be used without the need for expensive clinical trials for each new cancer type. Over-activation of the immune system and generation of autoimmune responses can occur but is manageable. Cancer immunotherapy is now a standard part of the modern anti-cancer armamentarium.**

